Final Exam Portfolio

ITAI 2374 – ROBOT OPERATING SYSTEMS & PLATFORM

SPRING 2026

EVA ABOU HARB

Module 0 – Entrance Questionnaire

Why I Took This Course?

- Learn how robot operating systems work
- Understand how AI is applied to robotics

Career Interests

- AI and Robotics
- Cybersecurity
- NLP

Degree Program

- Major: Artificial Intelligence and Robotics

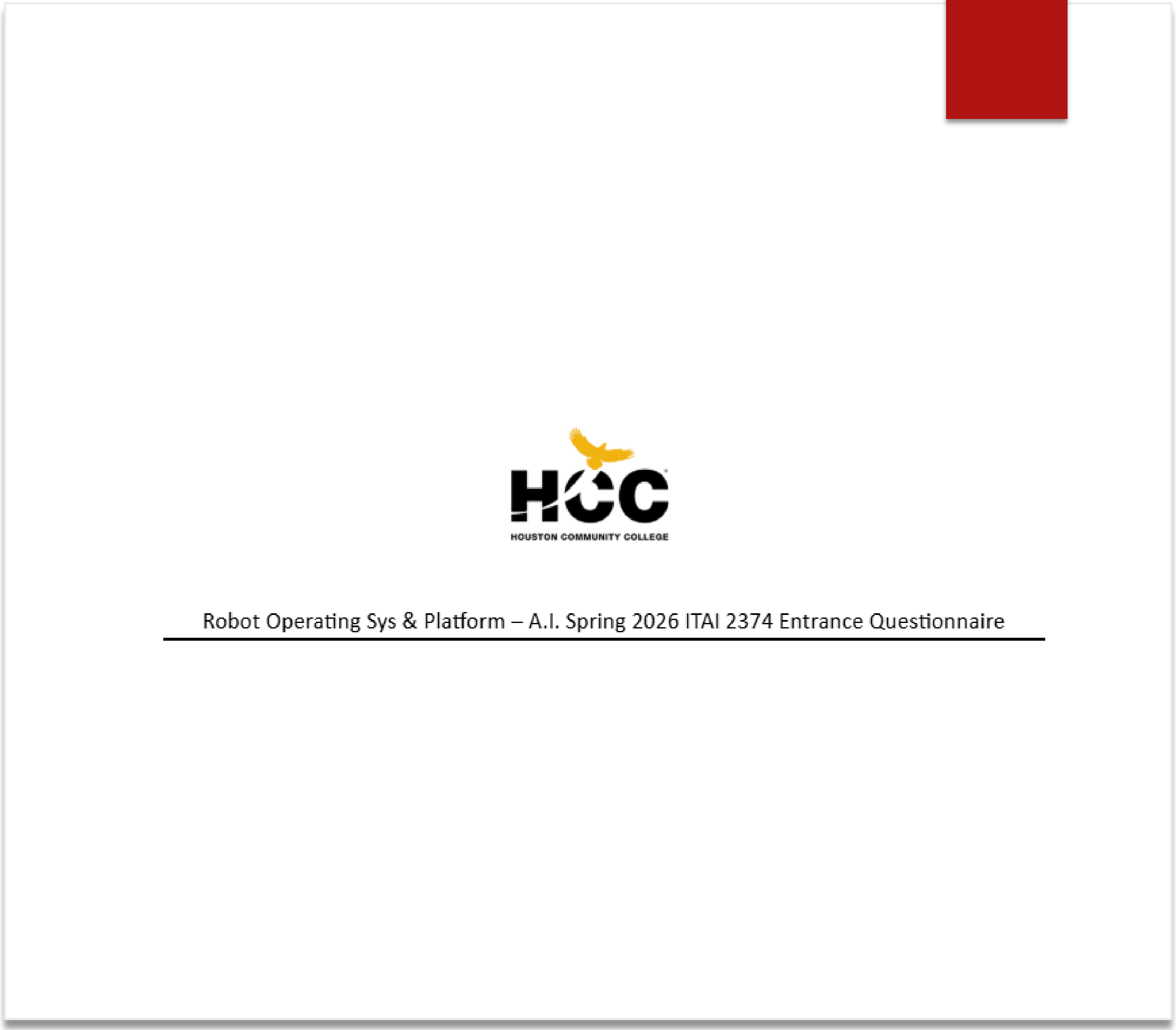
Reflection

- This assignment helped me define my career goals and understand how this course supports my future in robotics and intelligent systems.

Q00 Eva Abou Harb ITAI 2374 2026Spr

Full Entrance Questionnaire: View Here

- https://docs.google.com/document/d/11FjgLwl23ZtfWAkzeYlrXdfgGgodBd0m/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true


HOUSTON COMMUNITY COLLEGE

Robot Operating Sys & Platform – A.I. Spring 2026 ITAI 2374 Entrance Questionnaire

Module 01 - Team EVO Introduction

Team Name

- EVO

Team Motto

- "You can do it!"

Team Song

- **"Evolution"** by Sheryl Crow

Team Strength

- Adaptability

Communication Platform

- GroupMe

Team Members

- Eva
- Andre
- Muhammad

Reflection

- This assignment showed how teamwork, communication, and adaptability are important in Artificial Intelligence projects. Our team choices reflected how AI systems continuously learn, improve, and adapt to new challenges.

S01 EVO Eva Abou Abou Harb ITAI 2374 2026Spr

- https://docs.google.com/document/d/1grxD3uiqH1iKDJOA9MML36zo2A88ZQoE/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Module 01 – Raspberry Pi Research

Resource Used

- **Raspberry Pi Official Magazine – Issue #161 (January 2026)**
- **View PDF Resource Here:**
- https://drive.google.com/file/d/1czAIBEKIWtzbXYziADfSAvI19GEgsPGT/view?usp=sharing

Focus Area

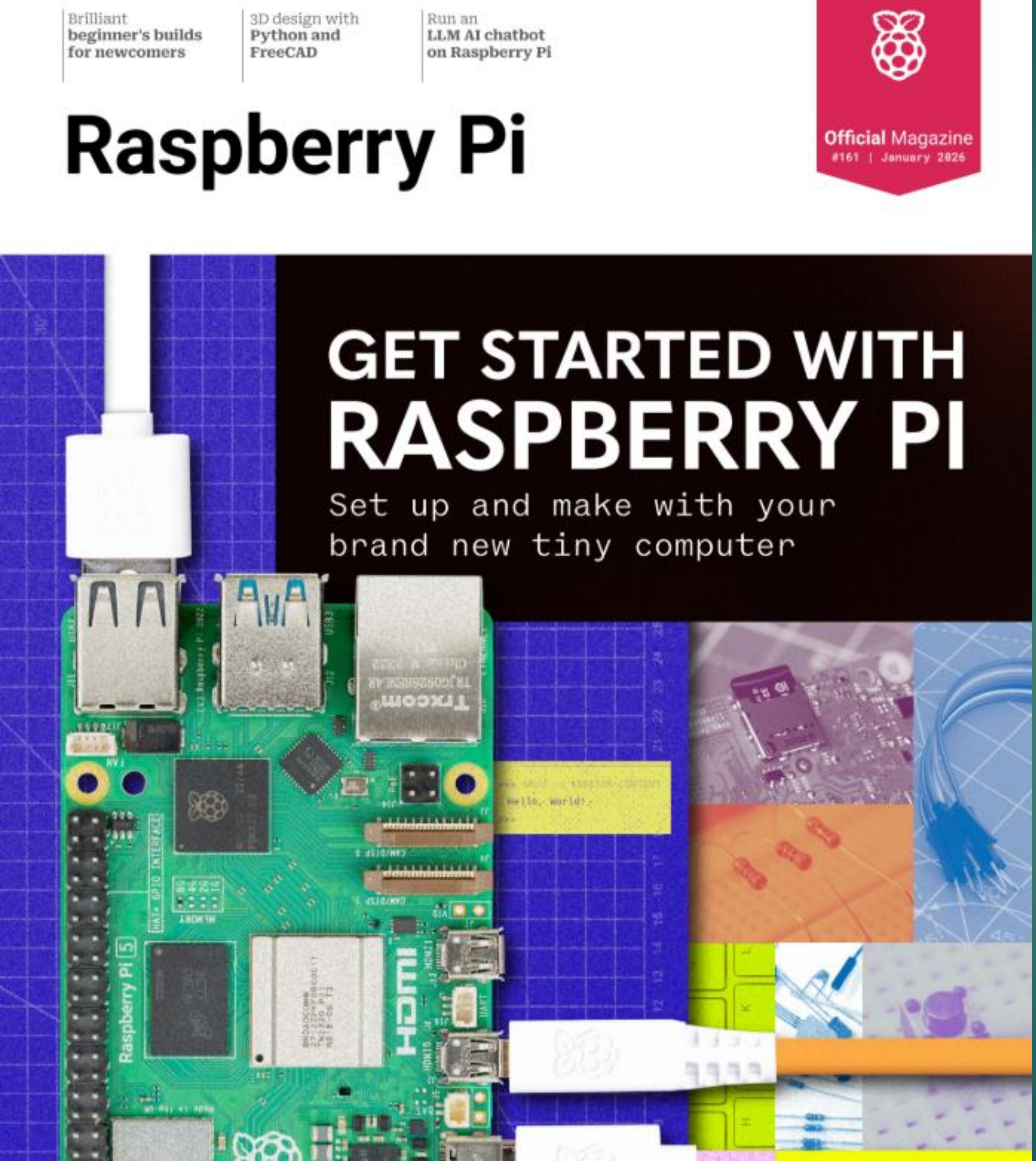
- **"Get Started with Raspberry Pi"**

What I Learned

- Raspberry Pi works as a complete computer
- It supports programming and electronics projects
- GPIO pins allow interaction with hardware
- Raspberry Pi OS is installed using a microSD card
- Raspberry Pi can even support local AI tools like LLM chatbots

Why it Matters

- This assignment helped me understand how hardware and software connect in robotics. Since my major is Artificial Intelligence and Robotics, learning how Raspberry Pi supports coding, automation, and sensors is very important for future projects.



A01 Team EVO ITAI 2374 2026Spr

View Full Raspberry Pi Research Here:

- https://docs.google.com/document/d/1PmWwmxowzNKyFqbETbS_ZbcvOEzwlpTx/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Module 04 – Creating an AI Video Agent with Synthesia

Objective

- Create an AI video agent showing what Houston Community College students can do with Artificial Intelligence.

Team Members

- Eva
- Andre
- Muhammad

What We Did

- Used Synthesia Free Tier
- Selected a template and AI avatar
- Created a professional AI-generated presentation
- Submitted a public working video link for streaming

What I learned

- As an AI and Robotics student, understanding how AI tools improve communication and automation is important for future careers in technology and business.

Module 04 – Creating an AI Video Agent with Synthesia

Course Reference Used

- **“Computing using the Raspberry Pi 004 – D”**
- This PowerPoint helped us identify the correct format for a working Synthesia public link. Slides 52 and 53 showed examples of valid links beginning with https://share.synthesia.io, which helped us verify our video submission.

View Course Reference Here

- https://drive.google.com/file/d/1osNz7pLwCG7h6K4k8Zs6QeHgcyMyJQyH/view?usp=sharing

A04 EVO Eva ITAI 2374 2026Spr

View Word Submission Here

- https://docs.google.com/document/d/1CX-_i9f50WLAjRBQwKQDwWmi_97Q-4ZH/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Watch Our Synthesia Video Here

- https://share.synthesia.io/39852129-9d92-4ef0-96cd-d11d4e011d26

Module 04 – Raspberry Pi Lab Setup and Hardware Testing

Team Members

- Eva
- Andre
- Muhammad

Activities Completed

- Installed Raspberry Pi OS using Raspberry Pi Imager
- Booted the Raspberry Pi successfully
- Connected monitor, keyboard, speaker, and camera
- Configured audio output using a JBL speaker
- Tested video feed using the Cheese application
- Installed and tested the Sense HAT module

What I Learned

- This lab helped me understand how hardware and software work together in embedded systems, I learned how Linux devices require proper setup, manual configuration, and troubleshooting for successful operation.

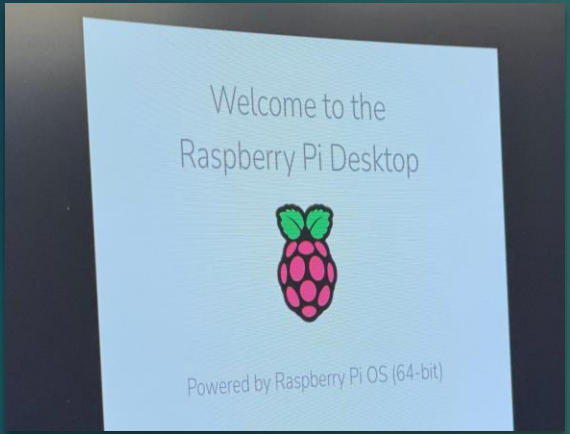
Biggest Challenge

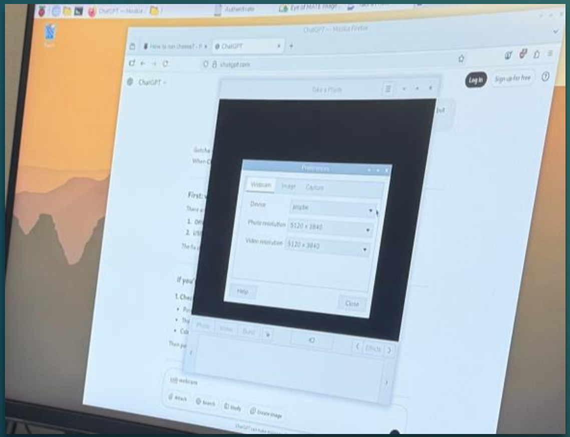
- The camera did not activate automatically and the Sense HAT was initially placed incorrectly. We solved this by manually selecting the camera device in Cheese and correctly aligning the GPIO pins for the Sense HAT.

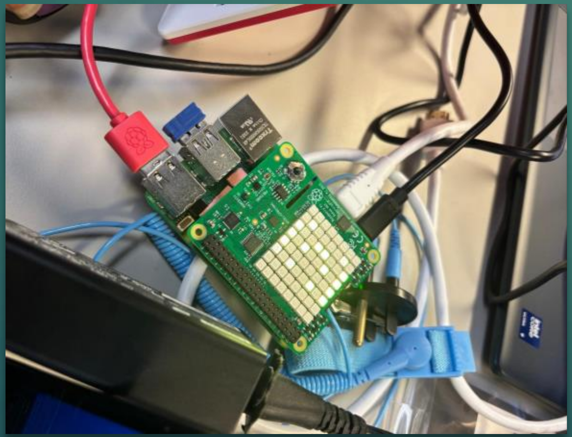
Why It Matters

- This experience is important for robotics because robots depend on sensors, cameras, and hardware communication. Learning how to configure these systems is essential for AI and robotics projects.

Module 04 - L04 Lab ITAI 2374







- ***Figure 1:*** Raspberry Pi OS desktop after boot
- ***Figure 2:*** Raspberry Pi hardware setup with connected peripherals
- ***Figure 3:*** Camera device settings in the Cheese application
- ***Figure 4:*** Correct Hardware Alignment and GPIO Pin Connection

Module 04 – L04 Raspberry Pi Technical Lab Report

Full Assignment

View Technical Report Here

- https://docs.google.com/document/d/1mhZaHFrexoG_mK0ZsUPcTScGgfBZIbnQ/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Module 04 – Intel Corporation Video Presentation

Objective

- Create a team video presentation about Intel Corporation and explain its importance in Artificial Intelligence, robotics, and future career opportunities.

Team Members

- Eva
- Andre
- Muhammad

What We Did

- Researched Intel's role in processors and AI technology
- Studied how Intel chips support robotics and edge AI devices
- Explored internship and career opportunities at Intel
- Created and submitted a team video presentation

What I Learned

- This assignment helped me understand how companies like Intel power robotics systems through processors, edge computing, and AI hardware. I also learned how industry leaders connect classroom knowledge to real-world careers and internships.

Why It Matters

- As an AI and Robotics student, understanding companies like Intel is important because robotics systems depend on powerful processors, sensors, and intelligent computing devices.

Intel04 EVO Eva ITAI 2374 2026Spr

View Word Submission Here

- https://docs.google.com/document/d/1J63pGw9FlaxckQ_pGRQcJNuVW2x0Bvwz/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Watch Our Intel Video Here

- https://share.synthesia.io/6ce74e0a-a446-461b-a43b-1ee1b93d28c0

Module 04 – P04 Class Presentation

Objective

- Present our AI Video Agent project to the class and explain how Houston Community College students can use Artificial Intelligence in professional environments.

Team Members

- Eva
- Andre
- Muhammad

What We Presented

- AI in daily life applications
- Smart assistants and automation tools
- AI in education and business communication
- AI-generated presentations using Synthesia
- How AI improves efficiency and problem-solving

What I Learned

- This presentation helped me understand that Artificial Intelligence is not only used in robotics or programming—it is also part of everyday life through smart devices, virtual assistants, recommendation systems, and workplace tools.

Why It Matters

- As an AI and Robotics student, understanding practical AI applications helps connect classroom learning to real-world technology. It also shows how AI impacts people beyond engineering fields.

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Watch Our Presentation Here

- https://share.synthesia.io/39852129-9d92-4ef0-96cd-d11d4e011d26

Module 05 – A05 NVIDIA GTC Conference Research

Objective

- Watch the full NVIDIA GTC 2026 keynote by Jensen Huang, and each team member selected a different recorded session to explore various topics in AI, robotics, and advanced computing.

Team Members

- Eva
- Andre
- Muhammad

My Session

- I focused on the session **Agentic AI 101**, which explained how AI is evolving from simple prompt-based tools into autonomous systems that can reason, use memory, and complete complex tasks independently.

What I Learned

- AI agents can work autonomously
- Multi-agent systems can collaborate together
- AI can solve real-world business problems
- Companies like ServiceNow use AI agents to resolve customer service tickets automatically
- AI is moving beyond traditional chatbots like OpenAI ChatGPT

Team Learning Highlights

- **Andre:** Jensen Huang keynote, CUDA, Blackwell systems, Vera Rubin architecture
- **Muhammad:** Robotics, physical AI, humanoid robots, digital twins, synthetic data

Why It Matters

- This assignment showed how AI is becoming more autonomous and intelligent. Watching the keynote and recorded sessions helped me understand where the future of robotics, automation, and smart systems is heading.

A05 EVO Eva ITAI 2374 2026Spr

Watch NVIDIA GTC Keynote Here

- https://www.youtube.com/live/jw_o0xr8MWU?si=S5LCeJOhdGejciyQ

View Assignment Here

- https://docs.google.com/document/d/18zwAH4CMcfNH2HXyHTYDEBYOSvGGshVm/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Module 05 – L05 SunFounder PiCar-X Robot Lab

Objective

- Configure and test the SunFounder PiCar-X robot using a Raspberry Pi 5 and verify its movement, steering, and hardware functionality through Python programming.

Team Members

- Eva
- Andre
- Muhammad

Activities Completed

- Enabled the I2C interface using **sudo raspi-config**
- Updated the operating system using Linux terminal commands
- Verified Python movement scripts
- Tested steering servos and drive motors
- Calibrated locomotion and obstacle avoidance
- Troubleshot camera and power issues

What I Learned

- This lab helped me understand how robotics depends on both hardware and software. I learned how Linux commands, Python scripts, GPIO communication, and motor calibration work together to make a robot move successfully.

Biggest Challenge

- The robot camera was non-functional, and we also faced battery power issues, keyboard problems, and command syntax errors. We solved this by focusing on locomotion, steering calibration, and debugging terminal commands.

Why It Matters

- This lab gave me real experience with robotics programming and system troubleshooting, which are essential skills for Artificial Intelligence and Robotics careers.

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View Technical Report Here

- https://docs.google.com/document/d/16b-JsROXB06wKbAcuXrEirtrzl9Jc7xN/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Watch Robot Movement Here

- https://youtube.com/shorts/uI3ReHcA_SI?si=Rq2M4gzP3CvSf16H

UR3e Cobot Setup and Pick-and-Place Programming

Objective

- Work with the Universal Robots UR3e arm cobot to learn movement control, waypoint programming, and simple pick-and-place automation.

Team Members

- Eva
- Andre
- Muhammad

Activities Completed

- Powered and Initialized the UR3e robot
- Used the Teach Pendant and PolyScope interface
- Configured the Hand-E gripper
- Set up Tool Center Point (TCP) and payload settings
- Practiced Home position and movement alignment
- Programmed waypoint-based motion for pick-and-place tasks

What I Learned

- This project helped me understand how industrial robots depend on precise programming and safety configuration. I learned how waypoints, TCP calibration, payload settings, and motion control work together for accurate robotic movement.

Biggest Challenge

- We had difficulties with TCP orientation setup and access permissions inside the system. We solved this by reviewing the training videos and using the operational model password to unlock programming features.

Why It Matters

- This experience introduced me to industrial robotics and automation systems used in real workplaces. It directly connects to my future goals in Artificial Intelligence and Robotics.

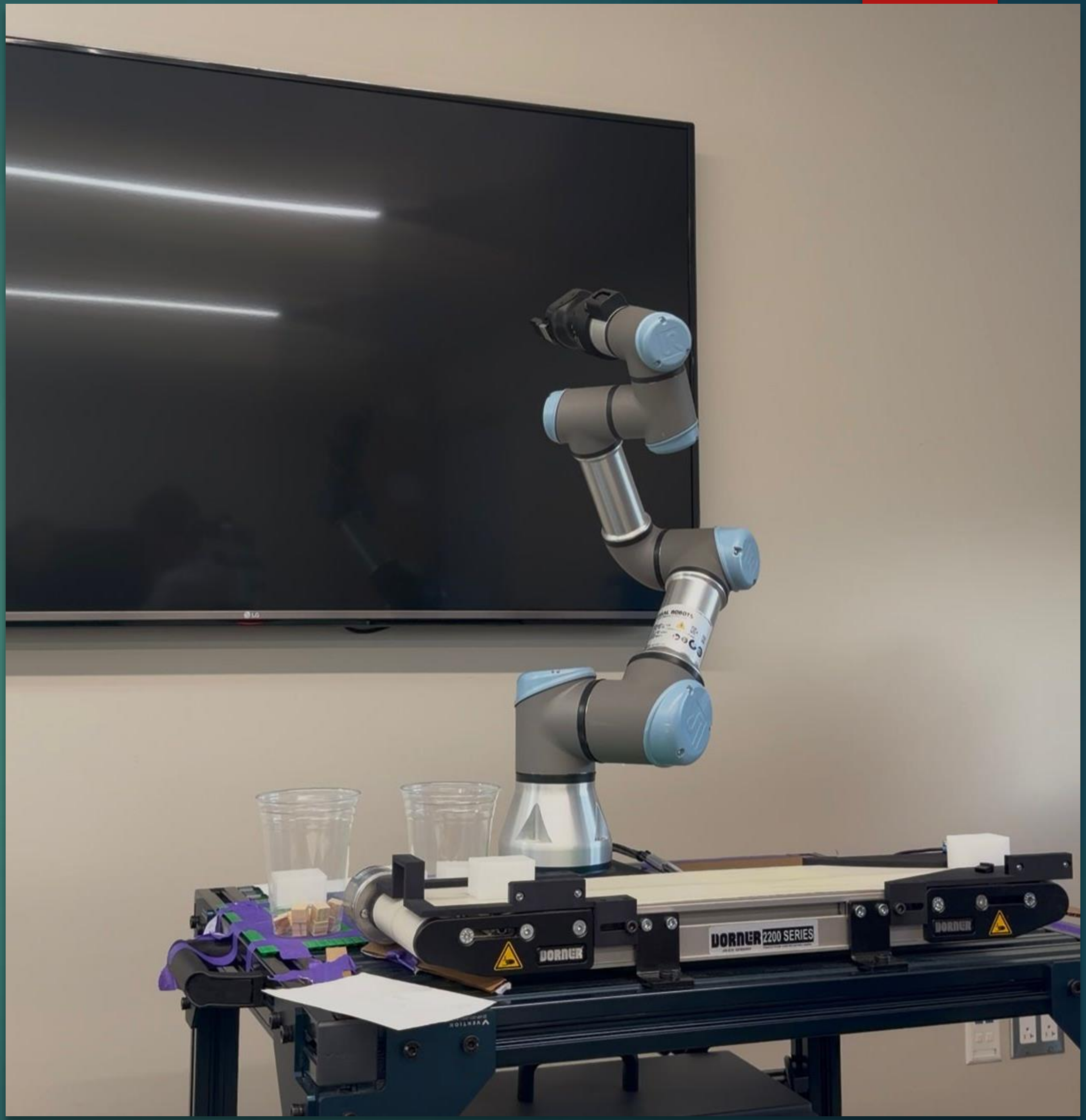
Module 06 – A06 EVO Eva ITAI 2374 2026Spr

View Assignment Here

- https://docs.google.com/document/d/1PxeZSyNeaDIJh5RgVtql6E0fqXBfMJd5/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Watch Pick-and-Place Demo Here

- https://youtube.com/shorts/oE1sZEHLwzY?si=ANAM9HoNtZ_4mVOG



Module 06 – P06 In-Class Presentation

Objective

- Present our team's progress with the Universal Robots UR3e robot and explain the development of our pick-and-place programming in the AI Lab.

Team Members

- Eva
- Andre
- Muhammad

What We Presented

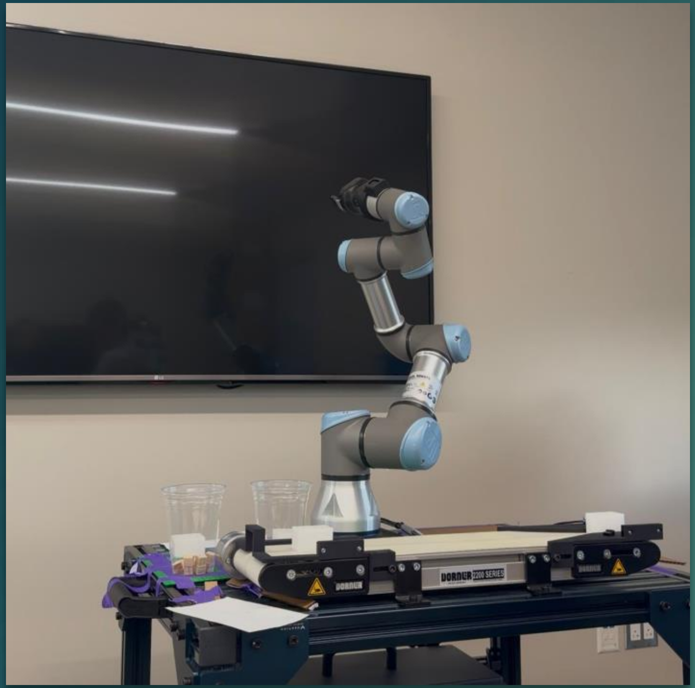
- UR3e robot setup and hardware configuration
- Waypoint programming and motion control
- TCP calibration and payload adjustments
- Pick-and-place workflow
- Challenges and troubleshooting process

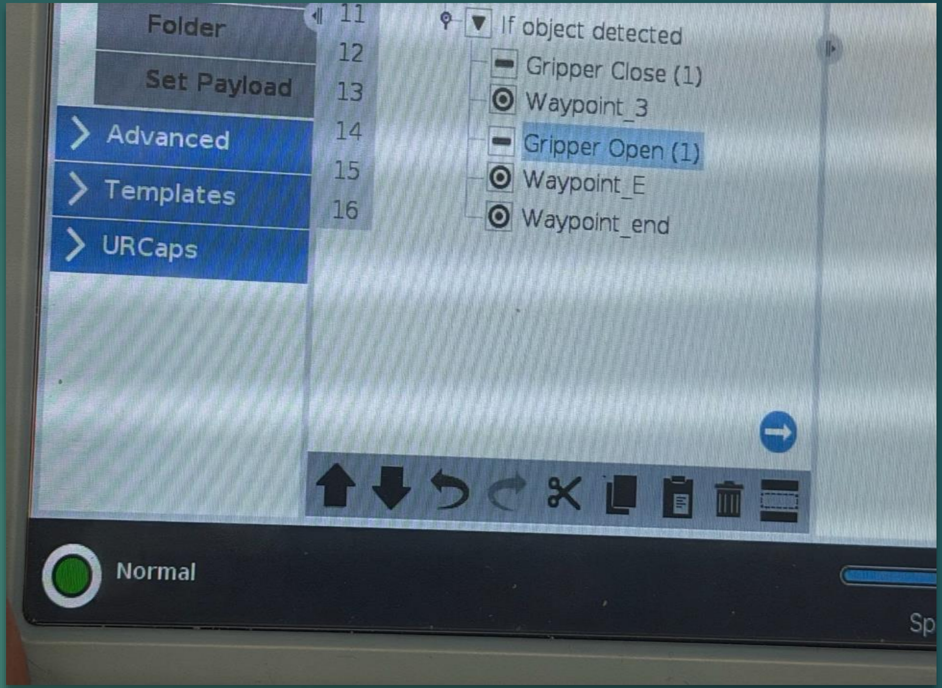
What I Learned

- This presentation helped me improve my confidence in explaining robotics projects clearly. I learned that communication is just as important as technical skills when working in engineering and automation.

Why It Matters

- In robotics careers, engineers must explain technical systems to teams, supervisors, and clients. This presentation strengthened my teamwork and professional communication skills.





Module 06 – UR3e Robot Progress Presentation

Participation Result

- Team presentation completed successfully during the April 28, 2026 AI Lab presentation.

Watch Our UR3e Pick-and-Place Demo Here

- https://youtube.com/shorts/oE1sZEHLwzY?si=ANAM9HoNtZ_4mVOG

Midterm Project – UR3e Cobot Pick-and-Place Automation

Objective

- Develop a complete automated pick-and-place system using the Universal Robots UR3e, Hand-E gripper, and Dorner 2200 Series conveyor.

Team Members

- Eva
- Andre
- Muhammad

Activities Completed

- Configured the UR3e cobot and Teach Pendant
- Calibrated TCP, payload, and center of gravity
- Programmed **MoveJ** and **MoveL** motion types
- Created advanced waypoint programming
- Added conditional logic using "If object detected"
- Built a full automated pick-and-place loop with conveyor interaction

What I Learned

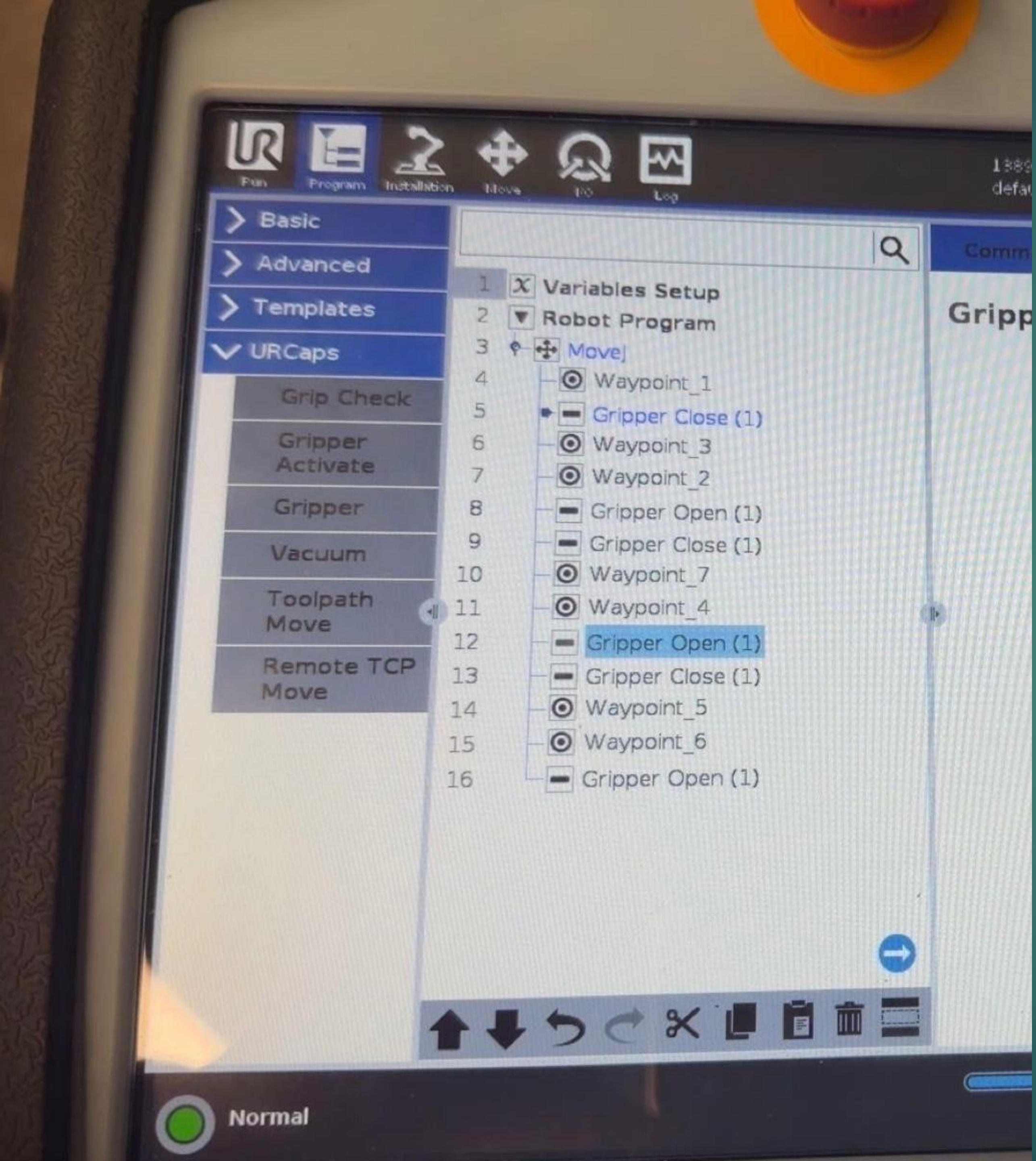
- This project taught me how industrial automation depends on both software logic and physical robot behavior. I learned that payload settings, object detection, and sensor calibration are critical for safe and accurate robot operation.

Biggest Challenge

- We experienced TCP misalignment, protective stops, gripper sensitivity issues, and access restrictions. We solved these by recalibrating the payload, adjusting object detection thresholds, and refining the TCP settings.

Why It Matters

- This was the strongest robotics project of the course because it introduced real industrial automation. It showed how robotics moves beyond simple movement into intelligent decision-making and full system integration.



Midterm Project Robotics

MT Project EVO Eva ITAI 2374 2026Spr

View Full Report Here

- https://docs.google.com/document/d/1RB7OX5O_awpgXN0VXNV5rKdnAMby7F-r/edit?usp=sharing&ouid=107180900369257048425&rtpof=true&sd=true

Watch Automated Pick-and-Place Demo Here

- https://youtu.be/w04tgsb4hGg?si=cBUa_ZuBsoLrvzWg

Final Reflection

What I Learned

- Throughout this course, I gained hands-on experience with Artificial Intelligence, robotics, and embedded systems through projects involving the Raspberry Pi, SunFounder PiCar-X, and the Universal Robots UR3e.
- I learned that robotics is not only about programming—it also requires troubleshooting hardware, configuring sensors, managing power systems, and working effectively as a team.

Strongest Area

- My strongest area was working with robotics labs, especially the UR3e cobot and the PiCar-X robot, because these projects connected directly to my major in Artificial Intelligence and Robotics.

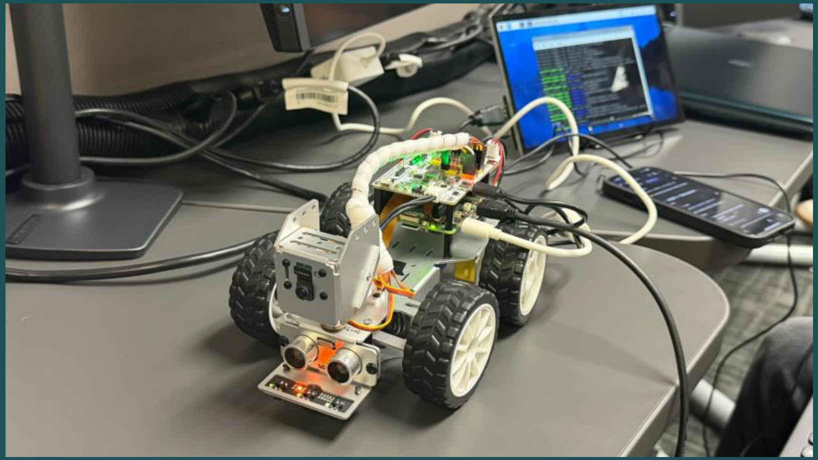
Biggest Challenge

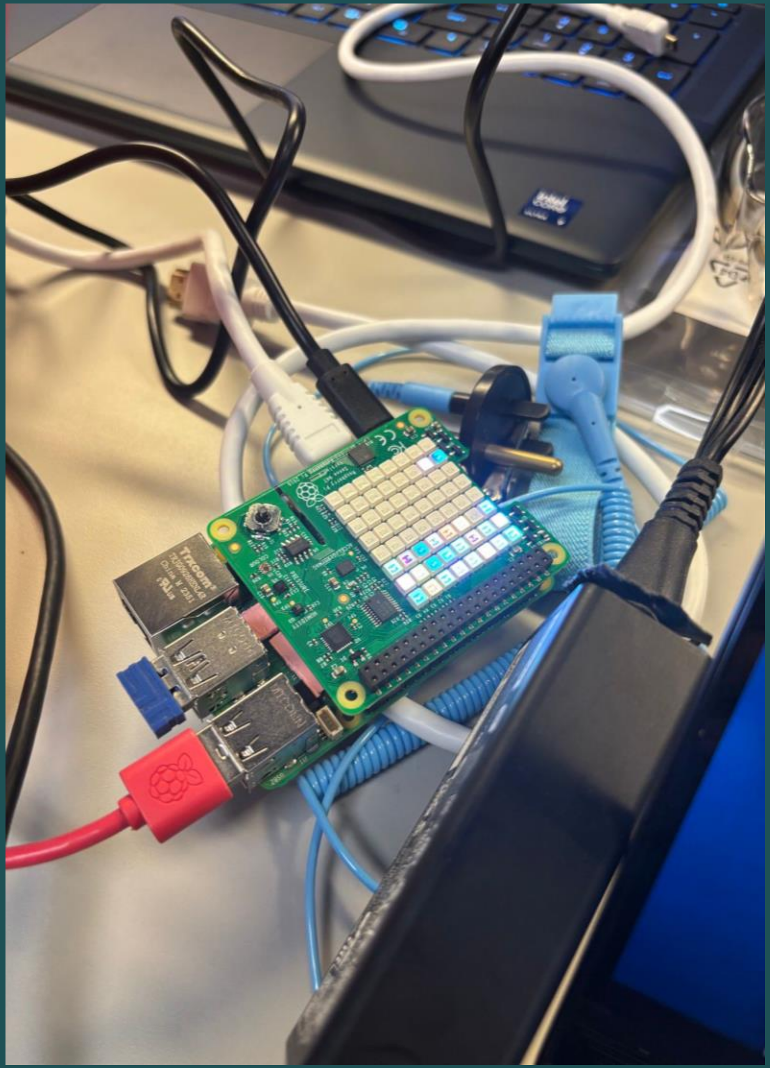
- My biggest challenge was solving hardware problems like camera failures, battery issues, and system configuration errors, but these experiences improved my patience and problem-solving skills.

Future Goal

- This course strengthened my interest in industrial robotics, automation, and intelligent systems, and it supports my future BAT degree in Artificial Intelligence and Robotics.

Course Highlights





"Small projects, big steps toward my future career."